INTELLIGENT DECISION SUPPORT SYSTEM FOR SRI LANKAN TEA ESTATE MANAGEMENT

INDIVIDUAL THESIS

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Tea Leaf Disease Detection & Pattern Analysis

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October 2026

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ABSTRACT

Tea growing is one of the significant agriculture activities in Sri Lanka. However, the tea estates still have challenges with early identification of leaf diseases. The manual examination process relies on people who may spend extra time examining and thus may miss some diseases or identify them later than needed. It makes the need for a rapid and accurate image-based tool for helping in disease management obvious. This research develops the Tea Leaf Disease Detection and Pattern Analysis component of the intelligent decision support system for managing Sri Lankan tea estates. The study concentrates on detecting and identifying leaf diseases using images by applying the YOLOv8s deep learning algorithm. For the model training and validation purposes, the authors use the tea leaf disease image dataset. After that, the researchers integrate the trained model with the web based system. This way, the developed component detects diseases, generates the confidence score, and gives suggestions on how to treat them. As a result, the validation results were: mAP@0.5 equal to 96.1%, precision of 95.2%, recall of 93.8%, and F1-score of 94.5%. It shows that the proposed model can provide efficient disease detection and helps with early leaf disease identification. The individual contribution to this project lies in designing and implementing the disease detection algorithm, integration of the model, its evaluation, and demonstration of the disease detection results within the overall Tea Guard decision support system.

Keywords: Tea Leaf Disease Detection, YOLOv8s, Deep Learning, Computer Vision, Tea Estate Management

ACKNOWLEDGEMENTS

It was indeed possible to complete this particular research work owing to the guidance and encouragement that was received during the course of the study. In this regard, sincere thanks are offered to the research supervisors, namely, **Ms. Uthpala Samarakoon and Ms. Suriyaa Kumari**, for the guidance and encouragement that was provided.

In addition, the author wishes to extend thanks to the **Faculty of Computing, Sri Lanka Institute of Information Technology (SLIIT)** for having provided the necessary academic environment and resources for conducting this research.

It is further extended to the members of R26-IT-118 who helped in developing the Intelligent Decision Support System as a whole for Sri Lankan Tea Estate Management. It is due to their help that the individual Tea Leaf Disease Detection and Pattern Analysis component could be integrated into the complete system.

It is also a matter of thanks to the people who were responsible for encouraging and giving feedback on the individual research component.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
DL	Deep Learning
F1-score	F1 Score
IoU	Intersection over Union
mAP	Mean Average Precision
ML	Machine Learning
RGB	Red, Green and Blue
YOLO	You Only Look Once
YOLOv8s	You Only Look Once version 8 small

1 INTRODUCTION

1.1 Overall project context

Tea is a vital agricultural product produced in Sri Lanka, and it is responsible for providing jobs, export earnings, and supporting rural communities. However, management of tea estates poses various challenges in terms of ensuring its sustainable operation. Challenges include labour management, diseases related to tea leaves, changes in weather, and maintenance of the quality of processed tea. Traditional tea estate management practices involve manual observation, record keeping, and reliance on experience of tea estate managers and field workers. Such a practice may not be very convenient in collecting and analyzing information in case of the necessity to make quick decisions across huge plantation areas. Fortunately, there are such innovations as Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), Internet of Things (IoT), and data analytics, which could facilitate this process.

The research project "Intelligent Decision Support System for Sri Lankan Tea Estate Management" was developed as a solution to some of the stated problems through integration of various technologies. In total, the overall system includes four main intelligent components: Labour Management and Welfare Optimization, Tea Leaf Disease Detection and Pattern Analysis, Tea Specific Weather and Climate Intelligence, and AI Based Tea Sorting and Grading Automation. While each of them focuses on a separate challenge in tea estate management, all the components provide common decision support system.

Overall system is supposed to provide useful information for tea estate managers through web based application. Integration of different information resources within one system should allow to make more prompt and informed decisions regarding tea estate management.

1.2 Individual component context

The individual component for this thesis is called Tea Leaf Disease Detection and Pattern Analysis. The presence of diseases in tea leaves is an important issue since

diseased tea leaves may impair the condition of the plant and its yield as well as the quality of the produced tea. Possible diseases include Anthracnose, Algal Leaf, Bird Eye Spot, Blister Blight, Grey Blight, and Red Rust among others.

Disease detection in many tea estates relies on visual inspection of leaves done either by a field worker or supervisor. The process heavily relies on the expertise of the person doing the job. Moreover, visual inspections can be complicated for a wide area which needs monitoring. Early symptoms can be overlooked, and variations between people's observations may cause inconsistency in disease identification.

The individual component aims at solving this problem using the image based deep learning approach to help detect and identify diseases in tea leaves. A user can upload a picture of a leaf to the system. The system will check whether the picture uploaded contains a tea leaf. Once a leaf is detected, the picture is forwarded to the disease detection model.

The model used is YOLOv8s trained on a tea leaf disease image dataset. The model will detect the disease and return the name of the disease and confidence score of the model. The system will also give advice regarding the treatment.

1.3 Problem statement

Tea leaf diseases are among the key problems that may affect tea estates' productivity and product quality. Diseases cannot be recognized at once, and therefore, they may spread throughout the entire estate affecting leaves and the overall quality of production. It is crucial to identify the diseases of tea leaves as soon as possible for efficient management of the estate.

Usually, the disease identification in tea estates is done manually with the help of field workers and supervisors. Such an approach requires human experience and takes time, especially in case of checking large plantation areas. The accuracy of the process may depend on the experience of the person who performs visual analysis. As a result, the final research paper identifies the disease of tea leaves as one of the key factors influencing tea estate productivity and efficiency.

In addition, manual inspection does not offer a fast and consistent way to analyze tea leaf images and identify the disease. Moreover, the manager or field worker also needs to make a decision about what kind of measures should be taken after the disease is identified. Therefore, disease identification is not always enough. The result of the detection should be associated with the information on the treatment that will help in making decisions.

The final research offers a solution to such a problem. An image based tea leaf disease detection component using the YOLOv8s deep learning model was created. The system allows uploading tea leaf image via the web application. Then the system processes the image and sends it to the YOLOv8s model that detects the disease of tea leaf, providing its name and confidence score. Based on that, the system recommends a treatment for the disease.

1.4 Research gap

There are previous studies demonstrating the usage of Artificial Intelligence and deep learning in disease detection from images. Image based deep learning approaches are used in many previous studies for disease identification from plant leaves. These approaches prove that automated disease detection could minimize the necessity of full manual inspections.

However, it can be seen that most of the studies examined in the literature review part of the research focus only on disease identification/classification without any further steps for connecting the results to actual tea estate management. It is clearly stated in the final research paper that many of the existing studies are focused on solving the separate problems instead of giving an integrated solution.

In the case of tea leaf disease detection, existing systems could detect the disease from the images. However, there is still a necessity to connect the result of the disease detection with some useful information about the management. Disease detection system becomes valuable if it gives the estate manager the disease detection result along with its probability of being detected and appropriate treatment recommendations.

The other gap is the lack of integration between disease detection and other tea estate management tasks. Almost all of the existing approaches are designed to solve only one agricultural problem. The Tea Guard system integrates the disease detection task with other tea estate management processes like labor management, weather intelligence, and tea grading.

Thus, the current research deals with the gap mentioned above by developing a tea leaf disease detection component using the YOLOv8s algorithm and integrating it into the web based intelligent decision support system. In addition to that, the developed component not only detects the disease but also gives a confidence score and treatment recommendations through the application.

The component becomes more practical for tea estate management because the result of the model becomes a base for the next step instead of just disease classification.

1.5 Aim and objectives

The aim of this research To develop and test an automated deep learning model capable of identifying diseases affecting tea leaves and recommending treatment methods for such diseases. The objective is to effectively control the disease problem in tea farms in Sri Lanka.

Objectives

- To study previous research and approaches to automated disease detection of tea leaves through image based Artificial Intelligence and deep learning.
- To collect and use the dataset of images of tea leaves for disease detection.
- To develop and train the deep learning YOLOv8s model for disease detection from tea leaf images.
- To evaluate the performance of the developed model through precision, recall, F1 score, and mAP@0.5.
- To integrate the developed YOLOv8s model into the Tea Guard web application in such a way that users will be able to upload their tea leaf images to receive disease detection results.
- To give a confidence score along with the detected disease and a treatment suggestion.

- To evaluate the usefulness of the developed component for Tea Guard decision support system.

1.6 Research questions or hypotheses

RQ1 - What are the common drawbacks of conventional techniques used to diagnose the disease in tea leaves of tea estates of Sri Lanka?

RQ2 - How efficient is the detection of tea leaf disease through images by the YOLOv8s deep learning technique?

RQ3 - In what ways can the developed disease detection algorithm using YOLOv8s be incorporated within the web based decision support system?

RQ4 - Is it possible for the developed system to provide information on the detected disease with accuracy along with a confidence score and a treatment recommendation?

RQ5 - How efficient is the performance of the disease detection algorithm developed by precision, recall, F1 score, and mAP@0.5?

Hypothesis

H1: An image detection system based on YOLOv8s can successfully detect the diseases of tea leaves through images and give useful information for tea estate disease management.

H2: An image detection system based on YOLOv8s cannot successfully detect the diseases of tea leaves through images and give useful information for tea estate disease management.

1.7 Scope limitations and assumptions

Scope

Tea Leaf Disease Detection and Pattern Analysis component of the overall Tea Guard Intelligent Decision Support System.

- Uploading tea leaf images through the web application.
- Processing the uploaded images.
- Detecting tea leaves and their diseases.

- Using the YOLOv8s deep learning model for disease detection.
- Providing the detected disease name.
- Providing a confidence score for the prediction.
- Providing a treatment recommendation.
- Storing the detection results in the system database.
- Displaying the results through the Tea Guard dashboard.
- Integrating the disease detection component with the wider Tea Guard decision-support platform.

Limitations

First of all, the performance of the disease detection model is dependent on the image quality and variability of tea leaf images used for training and testing. If there are poor-quality images that are hard to recognize or that differ from those used in training, the prediction might be affected.

Secondly, since the system is based on specific disease categories in the dataset, it can be assumed that the system can only detect diseases that are present in this dataset. Another factor that may influence the image-based disease detection is the environmental conditions like lighting, background, leaf orientation, image quality, and visibility of symptoms.

Another limitation of the system is the fact that the recommendation for the disease is not a replacement for an agricultural expert's opinion. The recommended treatment should be treated as a support and needs to be verified according to the agricultural knowledge before using it in practice.

Assumptions

The research assumes that the uploaded image contains a visible tea leaf and that the image quality is sufficient for the model to process. It is also assumed that the disease labels used in the dataset are suitable for training and evaluating the model.

The system also assumes that users upload images that are relevant to tea leaf disease detection. The treatment recommendations are assumed to be suitable as decision-support information for the corresponding detected disease.

1.8 Individual contribution

The thesis outlines the individual contribution of Miuranga W.A.R (IT22209320) towards the research project R26-IT-118 entitled "Intelligent Decision Support System for Sri Lankan Tea Estate Management." This research project comprises of four major modules of intelligence and individual contribution is carried out for the module of Tea Leaf Disease Detection and Pattern Analysis Module.

The individual contributions are outlined by the development of tea leaf disease detection module using artificial intelligence technology. It included the research on the tea leaf disease detection problem, review of relevant literature, the processing of tea leaf disease image dataset, and the design of the deep learning solution.

In the designed solution, YOLOv8s model is used instead of the CNN model that was originally proposed in the initial proposal. YOLOv8s model is designed and trained using tea leaf disease image dataset with the help of object detection performance evaluation measures.

The contribution of the individual also involves integration of the model with the web application. The workflow that was implemented involves the ability of the user to upload the tea leaf image using the React.js front end. This image is then sent to the backend which is developed using Node.js. It is further sent to the Hugging Face API where the detection of the disease occurs. The results are then sent back to the backend and stored in MongoDB to be shown on the dashboard.

A part of the contribution of the individual is the integration of the treatment recommendation feature. Once the disease is detected, the disease name, its confidence score, and the recommended treatment for it are provided by the system.

Evaluation Measure	Result
mAP@0.5	**96.1%**
Precision	**95.2%**
Recall	**93.8%**
F1-Score	**94.5%**

1.9 Thesis structure

Chapter 1 - Introduction

In this chapter, the overall Tea Guard research project and the specific tea leaf disease detection component of the project are introduced. The problem, research gap, objective, objectives, research questions, scope, limitations, assumptions, and individual contribution are explained in this chapter.

Chapter 2 - Literature Review

In this chapter, the literature review on tea leaf disease research, plant disease detection research, deep learning research, image-based disease detection research, YOLO-based object detection research, and agricultural AI system research is done. Also, the limitations of current researches are discussed and how this current research fills the gap is explained.

Chapter 3 - Research Methodology

In this chapter, the research methodology used to develop and evaluate the individual component is explained. The research approach, the dataset, data preparation, model development, model training, evaluation method, and research procedure are explained in this chapter.

Chapter 4 - Product and Solution Design

In this chapter, the design of the disease detection solution that is developed is explained. The system architecture, user interaction, image processing flow, YOLOv8s detection, backend interaction, database interaction, treatment recommendation, and Tea Guard integration are explained.

Chapter 5 – Implementation, Integration and Testing

In this chapter, the implementation of the proposed system is discussed. The chapter covers the technologies employed, YOLOv8s integration, frontend and backend implementation, Hugging Face API integration, database implementation, and system testing.

Chapter 6 – Results and Validation

The results of the proposed system are presented in this chapter. The performance of the disease detection model will be evaluated using precision, recall, F1-score, and mAP@0.5. In addition, the results of system testing and validation will be presented.

Chapter 7 – Discussion

The meaning of the obtained results is discussed in this chapter along with their comparison to the findings of previous studies. The advantages and disadvantages of the developed solution are also discussed in terms of its contribution to disease management at tea estates.

Chapter 8 – Commercialization and Sustainability

The practical and commercial applications of the developed system are considered in this chapter. The possible users of the system, its value for tea estates, and possible future use in the tea industry are discussed.

Chapter 9 – Conclusion and Future Work

Finally, the concluding chapter is about the presentation of the research results along with the discussion regarding how the research objectives have been met. In addition, the main contribution of this study is presented and improvement plans for the future, including the increase in the disease database, are provided.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This literature review was undertaken to explore current research, technologies, methodologies, and systems associated with detecting diseases in tea leaves and using artificial intelligence for agricultural decision-making support systems. Mainly the research focuses on image based plant disease detection, deep learning, convolutional neural network, object detection, analysis of tea diseases, treatment support, and implementation of artificial intelligence in agriculture decision making systems.

The review will also include studies that are related to the more general Tea Guard system as the management of tea estates is not just about disease recognition but also involves weather condition, crop productivity, tea quality, and many more. Thus, it is beneficial to understand the general environment of AI agriculture when designing a disease detection system which is part of the whole decision-making system.

The literature used in this chapter was collected from academic sources identified by the research proposal and final research paper. The studies included in the proposal were related to plant disease detection, tea disease classification, tea quality identification, agricultural deep learning, tea yield prediction, and climate based agriculture analysis. The final research paper further considered research related to labour management, tea disease detection, climate intelligence, and automated tea grading.

The review follows a focused narrative approach rather than claiming to be a systematic literature review. The main areas considered are

1. Artificial Intelligence in agriculture.
2. Deep learning and image-based agricultural analysis.
3. Convolutional Neural Networks for plant disease detection.
4. Tea leaf disease detection.
5. Object detection approaches for agricultural images.

6. Automated tea quality and grading systems.
7. Climate and environmental data analysis for tea management.
8. Integration of AI models with practical agricultural systems.
9. Limitations of existing systems.
10. The research gap addressed by the proposed Tea Guard component.

The scope is particularly focused on identifying how previous approaches can be improved from simple disease classification toward a more practical image-based detection and decision-support system for Sri Lankan tea estates.

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2.2 Domain and theoretical foundations

2.2.1 Tea Leaf Disease Detection

Tea leaf disease detection is one of the significant fields of agriculture technology as it has the potential to deteriorate the well-being, health, and productivity of the tea plant. Traditionally, the process of detecting disease is done manually by observing the visible symptoms on the leaves like spots, color changes, damage and pattern.

While manual detection is quite helpful, the accuracy of the process might depend on various factors like experience of the inspector, lighting condition, the quality of the image, and similarities between the symptoms of different diseases. When there is a large volume of data that needs to be inspected, the process might also be time-consuming. Therefore, computer vision and image analysis techniques can prove to be extremely helpful.

The individual research section of this project aims at Tea Leaf Disease Detection and Pattern Analysis. The system used in this research is based on tea leaf images as an input, and the deep learning-based object detection approach is used to detect the disease. YOLOv8s is used for tea leaf disease detection.

2.2.2 Artificial Intelligence in Plant Disease Detection

The application of Artificial Intelligence (AI) in agriculture for data analysis and decision-making has been steadily growing. The utilization of AI for plant disease identification can also be applied for analyzing images and detecting disease-specific patterns.

Deep learning algorithms can be especially efficient for image analysis tasks in agricultural applications since deep learning models can independently detect visual patterns based on training images. Such patterns can include color, texture, shapes, spots, lesions, and other visual patterns specific for diseases of plants.

The efficiency of deep learning algorithms for plant diseases detection was shown by Mohanty, Hughes, and Salathé [1], who demonstrated that deep learning models can detect disease-specific visual patterns in images of plants and use them for plant diseases recognition.

This finding represents a necessary theoretical background for the individual component since the Tea Guard disease detection system is also based on image-based deep learning.

2.2.3 Deep Learning for Tea Leaf Disease Detection

Deep learning is a type of machine learning algorithm that uses layered neural networks for learning from data. Deep learning models, in particular, can learn from the data without the need to define all the features associated with it. For instance, in image analysis, deep learning models can learn the required features directly from the images rather than having to define all visual features.

Tea leaf disease detection can thus be considered a good candidate for applying deep learning because various diseases can have various visual characteristics. A deep learning model can then learn from labeled tea leaves images, using the information to detect the disease on another image.

Tea leaf disease detection-related literature has also shown the application of deep learning in such tasks. Santhiya & Radhakrishnan developed TEADIS, a deep learning approach for tea leaf disease classification and segmentation based on field

observation data [2]. The study is relevant to the current research because it has applied deep learning in tea leaf disease detection.

The ultimate Tea Guard design involves the extension of this area through the application of YOLOv8s deep learning model in the application.

2.2.4 Computer Vision and Image Analysis

Computer vision enables computers to obtain useful information from digital images. In agricultural applications, computer vision can be used for plant identification, disease detection, crop monitoring, and quality assessment.

For tea leaf disease detection, the image is the primary source of information. The system analyses the visual content of the uploaded leaf image and identifies the disease category based on the patterns learned by the trained model.

The general process used by the individual component can be represented as

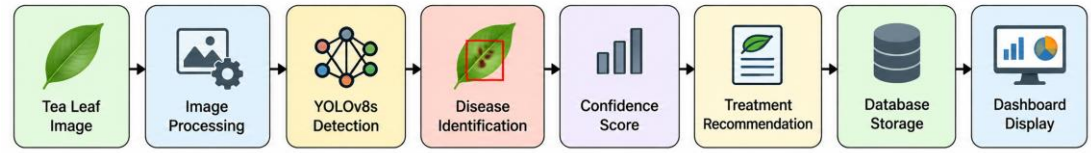


Figure 2.5: Tea Leaf Disease Detection and Treatment Recommendation Workflow

2.2.5 Object Detection

Object Detection is the process by which a certain region within the image is identified and classified accordingly. It differs from image classification in that, in object detection, the class label along with the position of the detected region is determined.

In case of plant diseases, object detection will be helpful in the situation where disease symptoms manifest in a certain part of the leaf. Rather than treating the entire image as a class, the model can recognize the relevant region and classify it accordingly.

The individual research component utilizes the YOLOv8s. This is an object detection model from the YOLO series. YOLOv8 comes from Ultralytics and is a contemporary computer vision model [3].

2.2.6 YOLOv8s

YOLO is short for You Only Look Once and means a group of models for object detection. These models are developed to detect objects and their locations in an image efficiently.

The final implementation of the component for detecting diseases on tea leaves uses the YOLOv8s model. This model was chosen as the final one instead of the CNN models considered initially in the proposal.

The implemented procedure includes the ability of a user to upload a tea leaf image via the React.js frontend interface. The image is sent to the Node.js backend, which communicates with Hugging Face API that contains the YOLOv8s model. Detected disease and confidence score are returned back to the application and shown to the user.

2.2.7 Precision, Recall, F1-Score and mAP

Model evaluation is a key stage in disease detection research. Each evaluation metric gives different information about the model's performance in detecting diseases.

Precision refers to the proportion of true disease cases among the disease cases detected by the model. High precision means that the model makes fewer false disease detections.

Recall indicates the effectiveness of the model in identifying the existing disease cases in the evaluation dataset. A high recall means that there are fewer false negatives.

F1-score represents the balance between precision and recall. This metric is suitable when accuracy and false negatives are of equal importance.

mAP@0.5 is one of the common object detection evaluation metrics. The evaluation involves classification performance and localization performance with Intersection over Union at 0.5.

Considering the individual Tea Leaf Disease Detection component, the resulting 96.1% mAP@0.5, 95.2% precision, 93.8% recall, and 94.5% F1-score prove the efficiency of the implemented YOLOv8s model on the evaluated tea leaf disease detection task.

2.2.8 Treatment Recommendation

While disease detection alone is helpful, the detected disease itself must carry some information for the user. Hence, the individual component contains disease-specific treatment recommendations.

Once the YOLOv8s network detects the disease, the output will show the disease name, confidence level, and its respective treatment recommendation. This would enable the system to deliver decision support information apart from just model output information.

This treatment recommendation is supposed to assist the user in knowing the management decision that could be done regarding the detected disease. This is a decision support tool and not meant to replace agriculture advice.

2.2.9 Web-Based Disease Detection

The individual component is implemented as part of a web-based application. The frontend is developed using React.js, while the backend uses Node.js.

The user uploads a tea leaf image through the web interface. The backend processes the request and communicates with the YOLOv8s model through the Hugging Face API. The prediction is returned to the backend and then presented through the user interface.

The system also stores disease detection results in MongoDB, allowing previous results to be retained for future reference.

Therefore, the individual component combines:

- React.js frontend
- Node.js backend

- Hugging Face API
- YOLOv8s object detection model
- MongoDB database
- Disease detection results
- Confidence scores
- Treatment recommendations

2.3 Related research

2.3.1 Deep Learning-Based Plant Disease Detection

One of the key studies related to image based plant disease detection was done by Mohanty, Hughes, and Salathé [1]. In their study, they examined the applicability of deep learning in identifying plant diseases using images. The result obtained from the study showed that deep learning can automatically extract visual features of different plant disease categories.

What makes this research crucial for this current research is the fact that it proved the applicability of image-based deep learning in automated plant disease identification. Nonetheless, the research was carried out using general datasets of plant diseases and not a Sri Lankan tea leaf disease detection system.

This current individual research specifically focuses on tea leaves and utilizes a tea disease detection model of YOLOv8s.

2.3.2 Tea Leaf Disease Classification and Segmentation

Santhiya and Radhakrishnan designed TEADIS as a system for tea leaf disease classification and segmentation based on deep learning [2].

The work is especially relevant since the authors concentrated on studying tea leaf diseases and used field observation datasets for their experiments.

This study shows that deep learning may be successfully used for tea diseases and the significance of field image datasets for developing disease detection systems in agriculture.

The current research is aimed at solving similar problems by implementing YOLOv8s as the detection model. However, the current paper focuses on implementing a web system for detecting diseases in tea leaves based on the uploaded images.

2.3.3 Image-Based Agricultural Disease Detection

Existing research into plant diseases demonstrates that image-based techniques can decrease the reliance on purely manual methods of disease identification [1]. Rather than having the user identify all symptoms individually by sight, the trained algorithm is able to compare the picture to the learned features within the training data set.

Nevertheless, images in agriculture can be difficult as the symptoms of disease can differ based on the disease stage, light, background, camera quality, and state of the leaves. This means that the training data set plays an important role when creating a disease detection model.

That is why the Tea Guard single component aims at testing the tea leaf disease detection model and measuring its efficiency via object detection metrics.

2.3.4 Importance of Field-Based Tea Disease Detection

Tea plantation will have various natural environmental variations that might not be available in the laboratory settings of the image. The leaves may be imaged in different lighting condition, background, angle, and degree of disease development.

The study by Santhiya and Radhakrishnan is related to this work since it takes into account field observation data for analysis of the tea leaf diseases [2]. It stresses the need for developing disease detection method that is suitable for real world environment.

This current study focuses on providing an applicable web based disease detection process where the user uploads a tea leaf image and gets the model generated result.

2.3.5 Research Contribution Compared with Previous Studies

Research studies have already demonstrated the capability of deep learning to help detect plant and tea leaf diseases [1], [2]. But the single component of Tea Guard is developed specifically for implementation of a practical process flow and not just model evaluation. The implemented component links the following steps:

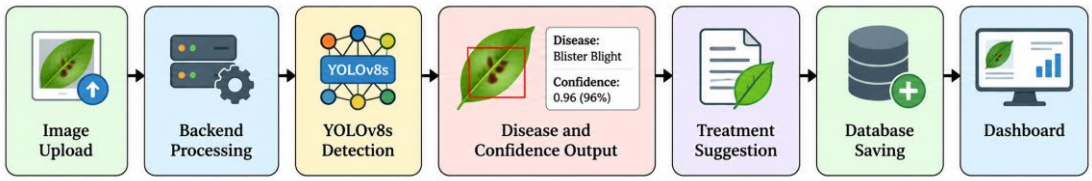


Figure 2.7: Integrated Tea Leaf Disease Detection and Decision-Support Process

2.3.6 Comparison of Related Research

Table 2.3: Comparison of Related Research Studies and the Proposed Component

Study	Main Area	Technology/Method	Main Contribution	Limitation Relevant to Tea Guard
Mohanty et al. [2]	Plant disease detection	Deep Learning/CNN	Automated plant disease recognition	General plant datasets
Santhiya & Radhakrishnan [3]	Tea disease	Deep Learning	Tea disease classification and segmentation	Mainly focused on disease analysis
Zhang et al. [4]	Tea quality	CNN/Transfer Learning	Automated tea quality identification	Focuses on quality rather than disease

Blasco et al. [5]	Agricultural grading	Machine Vision	Automated visual grading	Older feature-based approach
Mampitiya et al. [6]	Sri Lankan tea yield	Machine Learning	AI-based tea yield prediction	Does not perform image disease detection
Tea Guard	Tea disease + decision support	YOLOv8s + Web Application	Disease detection + confidence + treatment recommendation	Current research implementation

2.4 Existing systems products and techniques

2.4.1 Manual Tea Leaf Disease Identification

The conventional procedure of identifying tea leaves diseases is primarily based on manual visual detection. An inspector or an expert in agriculture examines the leaf visually and uses his prior experience for recognition.

This procedure works well if there is an experienced specialist available. But the problem arises when there is a need to examine a large number of leaves and diseases are visually similar to one another. The drawback of this procedure is that it does not create any digital record of the disease detected.

2.4.2 Image-Based Plant Disease Detection

Image-based disease detection systems use photographs of plants or leaves as input. A trained machine learning or deep learning model analyses the image and predicts the disease.

This approach has been investigated extensively in agricultural research. Mohanty et al. demonstrated the potential of deep learning for plant disease detection using images [1].

The main advantage of image-based detection is that the user does not need to manually identify all visual disease characteristics. Instead, the trained model performs the analysis and provides the predicted result.

2.4.3 Tea-Specific Disease Detection Systems

Tea-specific disease detection systems focus on diseases affecting tea plants rather than general agricultural diseases. TEADIS is an example of research that applies deep learning to tea leaf disease classification and segmentation [2].

Tea-specific systems are important because tea diseases have their own visual characteristics. A model trained on unrelated crops may not perform effectively when applied directly to tea leaves.

The Tea Guard individual component is also specifically designed for tea leaf disease detection and uses **YOLOv8s** as its final detection model.

2.4.4 Commercial Crop Disease Diagnosis Systems

Commercial agricultural applications have also introduced image-based disease diagnosis. Plantix is an example of a crop diagnosis platform that allows users to capture or upload crop images and receive information about possible crop problems and treatment-related guidance [4].

Such systems demonstrate that image-based disease diagnosis can be developed into a practical user-facing application.

However, a general commercial crop diagnosis platform is designed to support many crops and agricultural situations. The Tea Guard individual component is more specifically focused on tea leaf disease detection and is integrated into the research system developed for the Sri Lankan tea estate context.

2.4.5 CNN-Based Disease Classification

Convolutional Neural Networks have been widely used for image classification and plant disease recognition [1]. A CNN can learn visual features from training images and classify a new image into one of the learned categories.

CNN-based classification is useful when the main objective is to determine the disease category of the entire image. However, object detection provides additional localization information. The original project proposal considered CNN-based disease classification. However, the final implemented model is YOLOv8s. Therefore, YOLOv8s is the model that should be considered as the final technical approach in this thesis.

2.4.6 YOLOv8s Object Detection

YOLOv8 is a modern object detection model developed and distributed by Ultralytics [3]. The YOLO approach is designed to perform object detection efficiently and is suitable for applications where both detection and localization are required. The Tea Guard individual component uses **YOLOv8s** to detect tea leaf diseases. The model forms the core AI component of the system.

Table 2.4: YOLOv8s Object Detection Result

Evaluation Metric	**Result**
mAP@0.5	96.1%
Precision	95.2%
Recall	93.8%
F1-score	94.5%

2.4.7 Web-Based Implementation Technique

A major part of the individual contribution is connecting the disease detection model with a practical web application.

The system uses React.js for the frontend. The user can upload a tea leaf image through the interface. The image is sent to the Node.js backend, which handles the request and communicates with the Hugging Face API hosting the YOLOv8s model.

After processing, the system receives the disease prediction and confidence score. The result is then returned to the frontend and displayed to the user.

The detection result is also stored in MongoDB, allowing the system to maintain previous disease detection records.

2.4.8 Treatment Recommendation Technique

The system connects the detected disease with a treatment recommendation. This provides additional information to the user after the model produces its prediction.

The recommendation process can be represented as

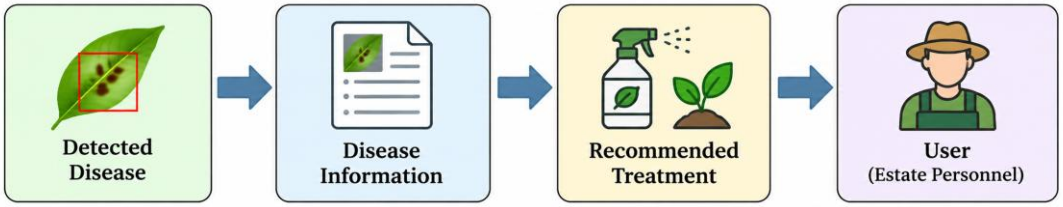


Figure 2.8: Tea Leaf Disease Treatment Recommendation Process

This approach makes the system more useful than a model that only returns a disease label. The purpose is to provide practical decision-support information based on the detected disease.

2.4.9 Comparison of Existing Techniques

Existing techniques for plant and tea disease detection can be broadly divided into manual inspection, CNN-based classification, object detection, and complete image-based applications.

Manual inspection depends on human knowledge and experience. CNN-based methods can automate disease classification [1]. Tea-specific research such as TEADIS demonstrates the application of deep learning to tea disease analysis [2]. YOLO-based object detection provides an alternative approach that can identify disease-related regions and classes [3].

The Tea Guard individual component combines object detection with a web-based interface, database storage, confidence reporting, and treatment recommendations.

Therefore, the research is focused not only on selecting a deep learning model but also on implementing a usable disease detection workflow.

2.5 Gap synthesis and conceptual framework

2.5.1 Research Gap

According to the literature review, deep learning has been successfully used to detect plant diseases based on images. Mohanty et al. have shown the potential of deep learning to develop an automated disease detector for plants based on images [1]. There have been studies related to classification and segmentation of tea leaf diseases, such as the TEADIS system [2].

This body of research serves as an excellent basis to develop an automated disease detector for tea leaves. Nevertheless, there is still a gap in creating a practical solution since a disease detector is only a part of the overall application and cannot serve alone. While the model is able to detect a disease, it is also necessary to develop a tool allowing to upload an image, get predictions, view confidence levels, and get treatment advice.

The Tea Leaf Disease Detection and Pattern Analysis module closes this gap by combining a YOLOv8s model with a web application.

2.5.2 Model and Implementation Gap

Previous research has involved various deep learning techniques used for the detection of plant and tea diseases [1], [2]. The original thesis research proposal involved the implementation of a CNN-based approach to detect tea leaf disease. But the model implemented in the final system is YOLOv8s for disease detection.

YOLOv8 is an object detection model that recognizes the relevant regions of the objects in an image along with the class they belong to [3]. This makes YOLOv8s an appropriate object detection technique that can be used for the Tea Leaf Disease Detection and Pattern Analysis model.

The performance of the final model was as follows -

- mAP@0.5: 96.1%
- Precision: 95.2%
- Recall: 93.8%
- F1-score: 94.5%

The final thesis concentrates on the implemented YOLOv8s model and not the CNN models proposed initially.

2.5.3 Practical Application Gap

Existing research demonstrates that deep learning models can identify plant diseases from images [1]. However, a machine learning model by itself is not a complete user-oriented solution. The model needs to be connected to an application through which users can provide images and understand the results.

The Tea Leaf Disease Detection and Pattern Analysis component addresses this practical requirement through a web-based implementation. The user can upload a tea leaf image using the React.js interface. The image is then sent to the Node.js backend, which communicates with the YOLOv8s model through the Hugging Face API.

After the image is analysed, the system returns the detected disease and confidence score. The result is then presented through the application and stored in MongoDB. This creates a practical connection between the deep learning model and the intended users of the system.

2.5.4 Decision-Support Gap

One additional drawback of the simple classification system is the provision of the disease name only. While disease detection is an essential step, the user might need to know the potential actions that could be taken upon detecting the disease.

In this regard, the Tea Leaf Disease Detection and Pattern Analysis module includes a Treatment Recommendation feature. The YOLOv8s model detects the disease and then the system gives not only the disease name and the confidence but also the corresponding treatment recommendation.

The aim of this feature is not to present the output of the machine learning algorithm only but to provide the information for making decisions.

2.5.5 Conceptual Framework

The conceptual framework for the Tea Leaf Disease Detection and Pattern Analysis module consists of four phases:

Firstly, in the input phase, the user submits the tea leaf picture using the web application.

Secondly, intelligent detection phase. In this phase, the submitted image undergoes processing at the Node.js backend server and is sent to the YOLOv8s model through Hugging Face API. This model analyses the image and detects the particular disease in the tea leaves.

Thirdly, the decision support phase. Disease detected in the previous phase can be used to give disease information and its treatment recommendation. Finally, output phase. In this phase, the disease name, the confidence of detection and treatment recommendation are sent back to the user as the detection output. The detection output is also saved in MongoDB.

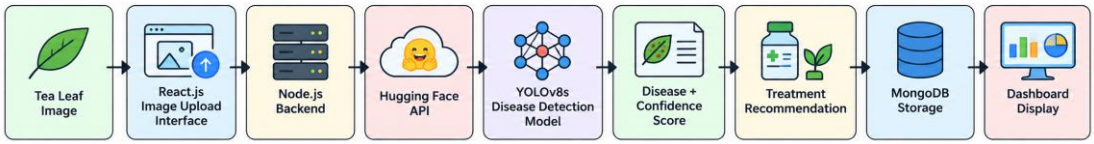


Figure 2.9: Conceptual Framework of the Tea Leaf Disease Detection and Pattern Analysis Component

2.5.6 Relationship Between the System Components

The React.js frontend provides the interface through which the user uploads a tea leaf image and views the detection result.

The Node.js backend manages communication between the frontend, YOLOv8s model, and MongoDB database.

The Hugging Face API provides access to the deployed YOLOv8s model used for disease detection.

The YOLOv8s model performs the main tea leaf disease detection task and produces the disease prediction with confidence information.

The treatment recommendation function provides relevant treatment-related information based on the detected disease.

The MongoDB database stores the disease detection results so that previous results can be retained.

The dashboard presents the final information to the user in an understandable format.

Therefore, these components work together to create a complete disease detection and decision-support workflow.

3 RESEARCH METHODOLOGY

3.1 Research design

The research follows a Design Science Research (DSR) and experimental development approach, which is suitable for designing, implementing, and evaluating an information technology artifact that addresses a real-world problem. The artifact developed in this research is the Tea Leaf Disease Detection and Pattern Analysis component of the Intelligent Decision Support System for Sri Lankan Tea Estate Management.

The research focuses on developing an image-based disease detection solution that can identify tea leaf diseases from uploaded images and provide useful treatment recommendations. The final implementation uses the YOLOv8s object detection model rather than the CNN-based approach initially considered during the proposal stage.

The research process consists of problem identification, dataset preparation, model development, model evaluation, system integration, and final validation. The main activities and their relationship to the thesis are shown in Table 3.1.

Table 3.1: Research activities and their relationship to the study

Research activity	**Activity in this study**	**Thesis chapter**
Problem identification	Difficulty of identifying tea leaf diseases accurately and providing timely treatment information	1, 2
Define objectives	Development and evaluation of an automated tea leaf disease detection system	1
Literature review	Review of deep learning, computer vision, tea disease detection and YOLO-based methods	2

Design and development	Dataset preparation, YOLOv8s model development and web application development	3, 4, 5
Demonstration	Working disease detection system with image upload and prediction	5, 6
Evaluation	Precision, recall, F1-score and mAP@0.5 evaluation	3, 6
Communication	Final thesis, system documentation and research findings	All

The development process is iterative. The first stage focuses on preparing the image dataset and developing the disease detection model. The second stage integrates the trained YOLOv8s model with the web application. The final stage focuses on testing the complete workflow from image upload to disease prediction, treatment recommendation, database storage and dashboard presentation.

3.2 Data participants or study materials

3.2.1 Tea Leaf Disease Image Dataset

The primary material that is utilized during this research is the dataset of pictures of tea leaf disease. This dataset includes pictures of tea leaves with various types of diseases and serves as the training data for the YOLOv8s object detection model.

These images supply the visual data necessary for learning the patterns of the diseases by the model. The dataset is prepared for the task of object detection in the way when the data about the disease are related to specific parts of the picture.

Finally, the implementation of the model involves the usage of the dataset to train and test the YOLOv8s model.

Table 3.2: Main study materials

Study material	Purpose
Tea leaf disease images	Training and evaluating the disease detection model
Disease annotations	Identifying disease regions/classes in images
YOLOv8s	Automated tea leaf disease detection
React.js	Image upload and user interface
Node.js	Backend processing and API communication
Hugging Face API	Model inference
MongoDB	Storing detection results
Treatment information	Providing treatment recommendations
Dashboard	Presenting detection results to users

3.2.2 Study Participants

The main disease detection experiment does not necessarily require the collection of any sensitive personal data about the participants. The data used in the main study is the tea leaf image database.

This system is designed for users like managers of tea estates, field officers, and others who are involved in tea production and disease detection. These users will have to interact with the system through the submission of tea leaf images and viewing of the disease detection and management results.

Any evaluation of the system by users in the future should be conducted independent of the model training experiment and should only capture information necessary for the evaluation of the system.

3.3 Experimental or study procedure

1. leaf image dataset preparation.
2. Image preprocessing and annotation preparation.
3. YOLOv8s model training.
4. Model validation and performance evaluation.

5. Web application integration.
6. Disease prediction and confidence-score generation.
7. Treatment recommendation.
8. Database storage and dashboard presentation.

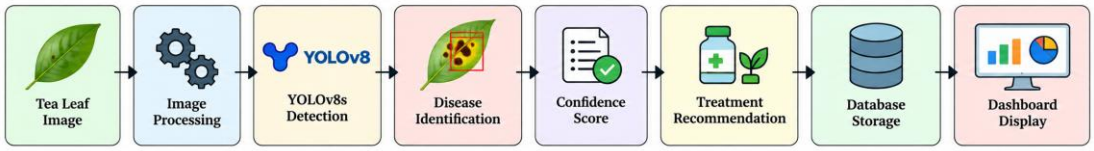


Figure 3.1: Experimental procedure for Tea Leaf Disease Detection and Pattern Analysis

3.3.1 Dataset Preparation

Data set preparation was the first step in carrying out the experiments. This entails the application of images of leaves having certain diseases. The images were sorted based on the various disease categories and the data set was pre-processed for object detection in YOLO format. The pre-processed data set was then divided into training, validation, and testing data sets.

3.3.2 Image Preprocessing

The gathered images were inspected and processed before being used for training the model. This includes inspecting the quality of the images, annotating the images, resizing where necessary, and arranging the data in the YOLOv8s dataset format. Some data augmentation methods were also used to improve the variety of images during training and assist the model in recognizing diseases even with different image conditions.

3.3.3 YOLOv8s Model Training

This prepared data set was employed for the training of the YOLOv8s object detection model. This model is able to learn the visual patterns related to the tea leaves' diseases and detect the areas affected by disease in the given input image. As a part of training, the parameters of the YOLOv8s model were tuned to enhance its detection capabilities.

3.3.4 Model Evaluation

Following the training of the model, the YOLOv8s model was evaluated based on common performance metrics used for object detection. Metrics such as precision, recall, F1 score, and mAP@0.5 were used to assess the performance of the model in the detection and classification of tea leaf disease. The final model had an accuracy of 95.2%, recall of 93.8%, F1-score of 94.5%, and mAP@0.5 of 96.1%.

3.3.5 Web Application Integration

Following model evaluation, the trained YOLOv8s model was integrated into the web-based application. The frontend was developed using React.js, while Node.js was used for backend processing. The backend communicates with the deployed YOLOv8s model through the Hugging Face API. When a user uploads a tea leaf image through the web application, the image is transferred from the frontend to the backend and then forwarded to the model for analysis.

3.3.6 Disease Detection and Output Generation

The uploaded tea leaf picture is fed to the YOLOv8s algorithm, which detects the disease from the picture. Along with the detected disease name, the confidence score for the detection is also provided by the model. The disease detection region can also be marked in the picture, giving the user an idea about the region that has been detected as infected.

3.3.7 Treatment Recommendation

Once the disease is identified, the system then fetches information regarding the management or treatment of that particular disease. Information provided will give additional information about the disease identified as well as the recommended management or treatment measures. The purpose of this module is to assist the tea estate users make faster decisions once they get the results of the disease identification.

3.3.8 Database Storage and Dashboard Display

The information on the final detection is kept in MongoDB for future use and analysis purposes. The information stored includes the detection and prediction related information. The results are presented using the web interface dashboard where the user can see both the current detection and previous detection information. This is how the whole process ends with the experiment from the beginning until the end.

3.4 Measures and evaluation metrics

The accuracy of the Tea Leaf Disease Detection and Pattern Analysis module was measured with the help of the following object detection metrics. The chosen set of metrics included Precision, Recall, F1-score, and mAP@0.5. They were used for assessing the ability of the YOLOv8s algorithm to detect tea leaf diseases.

Table 3.4: Model Evaluation Results

Metric	How measured	Target
Precision	Measures the proportion of predicted disease detections that are correctly identified.	High precision with fewer false-positive detections
Recall	Measures the proportion of actual disease cases correctly detected by the model.	High recall with fewer missed disease cases
F1-score	Measures the balance between Precision	High and balanced

	and Recall.	performance
mAP@0.5	Measures the overall object detection performance using an IoU threshold of 0.5.	High detection and localization performance

3.5 Data analysis

The collected image data on tea leaf disease were analyzed using the YOLOv8s model that had been trained beforehand. The analysis included assessing the performance of the model in detecting and recognizing disease regions on tea leaf images. The performance of the model was analyzed in terms of Precision, Recall, F1-score, and mAP@0.5.

The results were compared to evaluate the general performance of the model. Besides the analysis of the model, the entire workflow of the system was tested starting from uploading images, through disease detection, assigning confidence scores, making recommendations on treatment, storing results in a database, and showing results on a dashboard.

The analysis proved that the YOLOv8s model demonstrated 95.2% Precision, 93.8% Recall, 94.5% F1-score, and 96.1% mAP@0.5.

3.6 Ethics privacy security and responsible AI

There was an analysis of ethical, privacy, security, and responsible AI aspects in relation to the development of the Tea Leaf Disease Detection and Pattern Analysis part.

- **Privacy**: Only necessary information that is relevant to disease detection and the operation of the system is regarded. No unnecessary personal information is gathered or stored.
- **Data Security**: The detection results and the information stored in the system are secured by means of the proper database and application security mechanisms. Access to the system and the information stored there must be restricted to only authorized persons.

- **Responsible AI**: The disease predictions along with the certainty scores generated using the YOLOv8s model can be used as the decision-making information in order to detect possible diseases of the tea leaves. They are to be viewed as such and not as a definitive agricultural diagnosis.
- **Treatment Recommendations**: The treatment suggestions are offered as the decision-support information based on the disease detection. The proper agricultural recommendations should be considered before the treatment is applied, especially in case of chemical treatment methods.
- **Human Decision-Making:** The system is designed to assist tea estate users in making faster and more informed decisions. It is not intended to completely replace experienced agricultural officers or human decision-making.

3.7 Reproducibility and research artefacts

The primary research artefacts have been retained to ensure reproducibility of the Tea Leaf Disease Detection and Pattern Analysis component. These artefacts consist of the tea leaf disease images data set, the annotated images, YOLOv8s model, model evaluation artefacts and the source code used in the implementation of the system.

The web application artefacts are the React.js front-end, Node.js back-end, Hugging Face API, MongoDB database, treatment recommendation information and dashboard implementation. The primary system workflow and architecture diagrams have also been retained as research artefacts.

The process of dataset preparation, model configuration, model evaluation results and implementation of the system is explained in the research work. Retention of these artefacts enables reproduction of the experimental procedure and system development process through use of the same or an appropriately prepared dataset and implementation configuration.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

4.1.1 Stakeholders

Tea Leaf Disease Detection and Pattern Analysis is meant to help in making decisions on the diagnosis and treatment of diseases in tea estates. Some of the people who engage with the component are listed below, together with their interest in the component (Table 4.1).

Table 4.1: Stakeholders of the Tea Leaf Disease Detection and Pattern Analysis Component

Stakeholder	Role and Interest
Tea Estate Managers	Upload tea leaf images, review disease detection results, and use treatment recommendations to support estate management decisions.
Field Officers and Supervisors	Use disease detection results to identify possible tea leaf diseases in the field and support disease monitoring activities.
Tea Cultivators and Field Workers	Benefit from earlier identification of possible disease problems and access to relevant disease-management information.
System Administrators	Maintain the web application, manage system access, and support the reliable operation of the component.
Research and Development Team	Develop, evaluate, maintain, and improve the YOLOv8s model and its integration with the web application.

The main users of the component include tea estate managers and field officers since they will be able to use the component to analyze their disease forecasts and information on how to treat these problems.

The system was developed to help the user identify potential leaf diseases in tea. The prediction and treatment advice provided by the system must be taken as decision-support information.

4.1.2 Functional Requirements

Functional requirements describe the main operations that the Tea Leaf Disease Detection and Pattern Analysis component must perform.

Table 4.2 : Functional Requirements

Functional Requirement	**Description**
Tea Leaf Image Upload	The system shall allow users to upload tea leaf images through the web application.
Image Processing	The system shall send uploaded images to the backend for processing.
Disease Detection	The system shall use the YOLOv8s model to detect and identify tea leaf diseases.
Confidence Score	The system shall display the confidence score associated with the disease prediction.
Treatment Recommendation	The system shall provide relevant treatment and disease-management information based on the detected disease.
Result Storage	The system shall store disease detection results and related information in MongoDB.
Dashboard Display	The system shall display detection results, confidence scores, and treatment recommendations through the dashboard.

Detection History	The system shall support access to previously stored detection records where this functionality is implemented.

4.1.3 Non-Functional Requirements

Non-functional requirements describe the expected quality, security, usability, and operational characteristics of the component.

Table 4.3: Non-Functional Requirements

Non-Functional Requirement	**Description**
Performance	The system should process uploaded images and return results within a reasonable time.
Usability	The interface should be simple and easy to use for uploading images and understanding detection results.
Reliability	The system should handle image-processing requests and display results consistently.
Security	API credentials and database access should be protected against unauthorized access.
Maintainability	The system components should be organized to support future maintenance and updates.
Scalability	The component should support future improvements, including additional disease classes and model updates.
Data Integrity	Detection results stored in MongoDB should remain consistent and correctly associated with their relevant records.

4.1.4 User scenarios

User scenarios are descriptions of how the users will use the Tea Leaf Disease Detection and Pattern Analysis sub-system for disease detection and retrieving related treatment data. The primary scenario is when the tea estate manager or field officer will upload a tea leaf image and evaluate the predicted output from the YOLOv8s algorithm.

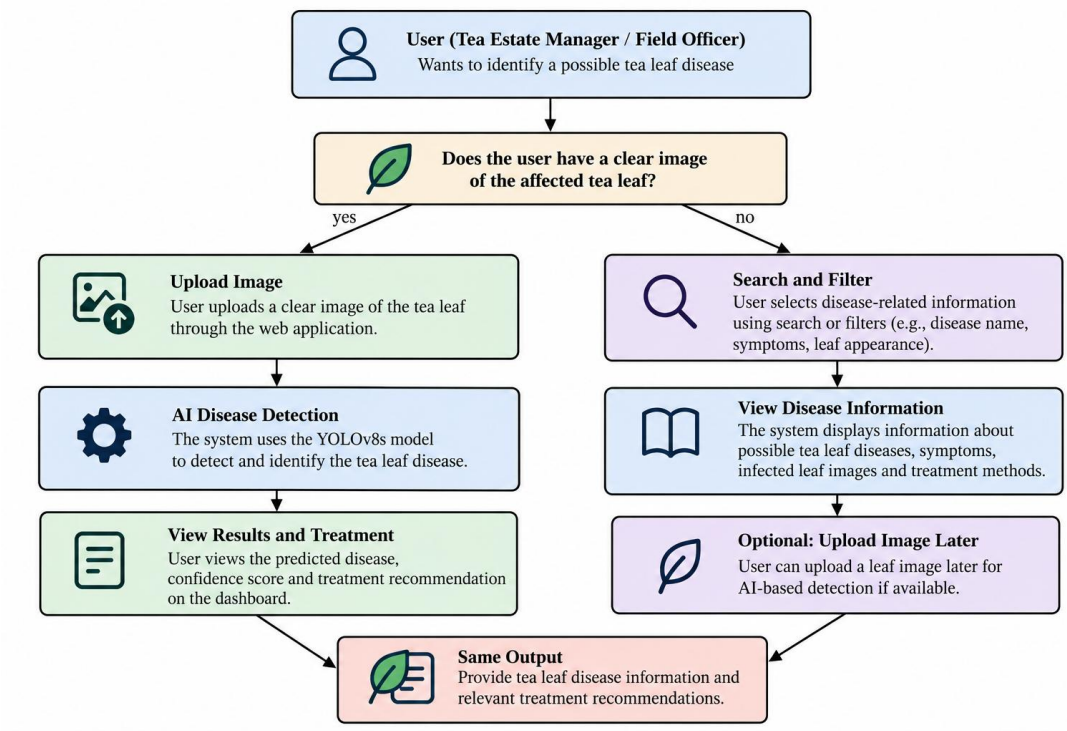


Figure 4.2: User Scenario for Tea Leaf Disease Detection and Pattern Analysis

Table 4.3: User Scenarios

Scenario ID	User	Scenario	Expected Outcome
US01	Tea Estate Manager	Uploads a tea leaf image through the web application.	The system accepts the image for disease detection.

US02	Field Officer	Submits an image of a potentially diseased tea leaf.	The YOLOv8s model identifies the predicted disease and provides a confidence score.
US03	Tea Estate Manager	Reviews the disease detection result.	The dashboard displays the predicted disease and relevant detection information.
US04	Field Officer	Checks the treatment recommendation associated with the detected disease.	The system displays relevant treatment and disease-management.
US05	Tea Estate Manager	Reviews previously stored detection results, where supported by the application.	The user can refer to available historical detection records.
US06	System Administrator	Maintains the application and its supporting services.	The component remains manageable and can be maintained when necessary.

4.2 Component architecture within the complete solution

The Intelligent Decision Support System for Sri Lankan Tea Estate Management uses the client server architectural structure based on the web platform. The total system includes four major parts: Labor Management and Welfare Optimization, Tea Leaf Disease Detection and Pattern Analysis, Tea Specific Weather and Climate Intelligence, and Automated Tea Sorting and Grading with the Use of AI. Each part has specific functionality and produces relevant results for the whole system.

The Tea Leaf Disease Detection and Pattern Analysis part is responsible for detecting the diseases of tea leaves using uploaded photos and recommending proper measures. It interacts with the major system through the web frontend, backend services, AI model API, and database.

Main architectural components

- Frontend (React.js): Provides the user interface for uploading tea leaf images and viewing disease detection results.
- Backend (Node.js): Receives image upload requests, manages application processing, and communicates with the AI model and database.
- AI model (YOLOv8s): Detects and identifies possible tea leaf diseases and returns prediction information, including confidence scores.
- Hugging Face API: Provides access to the deployed YOLOv8s model for image analysis.
- Database (MongoDB): Stores detection results and related information.
- Dashboard: Presents the disease prediction, confidence score, and treatment recommendation to the user.

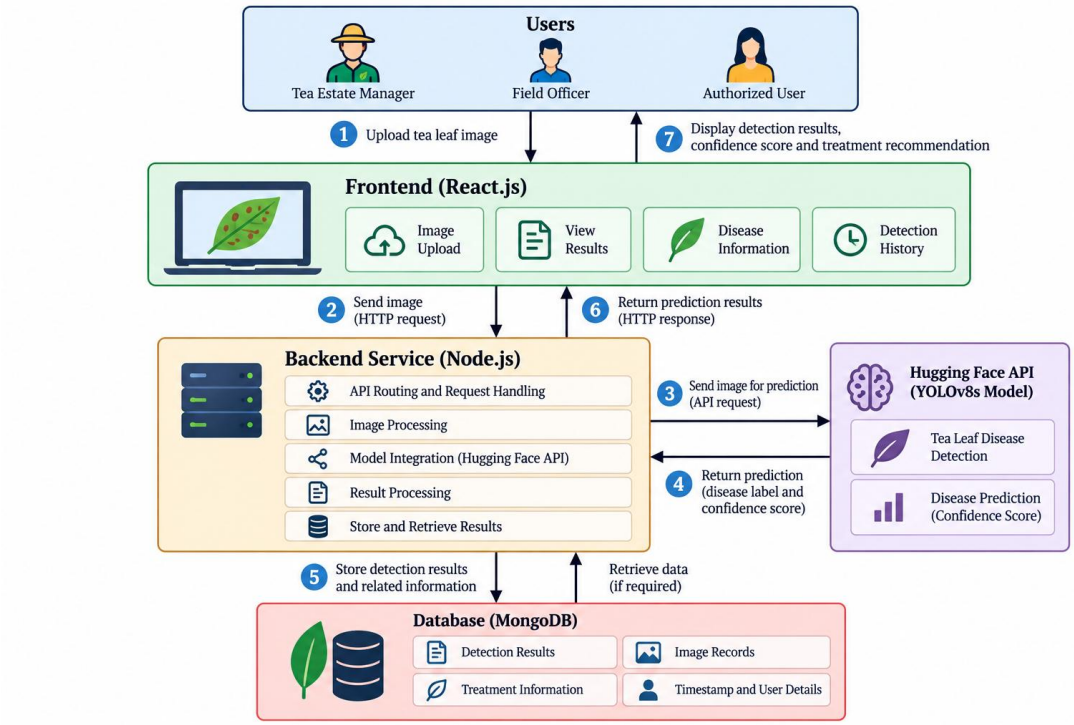


Figure 4.4: Component Architecture of Tea Leaf Disease Detection and Pattern Analysis

4.3 Design alternatives and decisions

There are several design options that were examined during the development of the Tea Leaf Disease Detection and Pattern Analysis system. The key considerations included choosing a disease detection algorithm, frontend, backend, technique for integrating the model into the software, and a database. Selected technologies allow for disease detection based on images and results presentation using a web application.

Table 4.4: Design Alternatives and Decisions

Design Aspect	Alternative	Selected Decision	Reason
Disease detection model	CNN-based image classification	YOLOv8s object detection	Supports disease identification and detection of affected regions in tea leaf imag

Frontend development	Other frontend frameworks	React.js	Supports a component-based interface for image uploading and displaying detection results.
Backend development	Processing requests entirely in the frontend	Node.js	Handles application requests and connects the frontend with the model API and database.
Model deployment and integration	Integrating model inference directly into the main backend	Hugging Face API	Provides access to the deployed YOLOv8s model through API requests.
Database	Relational database	MongoDB	Provides document-based storage for detection results and related information.
Result presentation	Displaying only the predicted disease name	Displaying the disease name, confidence score, and treatment recommendation	Provides more useful information to support disease-management decisions.

The choice of the YOLOv8s model for the final implementation was made since the component uses object detection instead of classifying the whole image by its general category. The model is able to make disease predictions and give confidence scores that can be further used in the web application for presenting the detection results.

React.js was chosen as a frontend framework to make the interface interactive, while Node.js controls the backend and interactions with other services. The Hugging Face API is needed to connect the backend with the deployed model, so that the inference

step doesn't need to be implemented on the frontend side. MongoDB was chosen as a database for storing the detection results and other relevant information.

4.4 Detailed design

The design of the Tea Leaf Disease Detection and Pattern Analysis module provides an explanation of how the frontend, backend, YOLOv8s model, and database collaborate to achieve disease detection in tea leaves and recommend appropriate treatment measures. This process follows an order where the user first uploads an image of a tea leaf, then the model analyzes the image and returns the results to the web application.

4.4.1 Frontend Design

The frontend is implemented using React.js. It serves as the graphical user interface for uploading images of the tea leaves and displaying the output produced by the disease detection algorithm. It displays the predicted disease, confidence level, and corresponding treatment advice in a readable manner.

The key functionalities of the frontend include uploading images, submission of images for analysis, and displaying the output.

4.4.2 Backend Design

The backend uses Node.js for development and acts as a mediator for communication among the frontend, AI model, and the database. After receiving the picture from the user, the backend will process it as needed and pass the image to the YOLOv8s model via the Hugging Face API.

After receiving the prediction, the backend processes the information received and manages the database actions.

4.4.3 AI Model Design

The function leverages the YOLOv8s model for recognizing potential diseases affecting tea leaves using uploaded images. It is able to make an analysis of the image and then provide the predicted disease and the confidence of that prediction.

The model is deployed via Hugging Face API in order not to host the model on the frontend side of the application.

4.4.4 Database Design

MongoDB is employed for storing the results of the detection process along with related details. The primary details that could be stored within a detection record include

- Disease name predicted
- Confidence score
- Pointer to the image that was uploaded
- Treatment recommendation
- Timestamp of the detection process

Depending upon the designed schema of the database, different fields could be included within a detection record. The stored information can be retrieved by the backend whenever needed.

4.4.5 Treatment Recommendation Design

Once a disease is identified, then relevant information on how to treat the disease will be provided based on the disease predicted by the model. This gives the user an opportunity to look at the recommendation along with the disease identified.

The recommendations are meant to aid tea plantation owners in making decisions about managing their diseases but not necessarily to diagnose the disease or give farming recommendations.

4.5 Standards and professional practice

The design of the Tea Leaf Disease Detection and Pattern Analysis feature takes into account appropriate software engineering considerations, security concerns, and proper use of artificial intelligence.

- API Security: Use secure communication and protect API credentials.
- Open-Source Licensing: Follow the license terms of the software libraries used.
- Data Privacy: Avoid collecting unnecessary personal information and protect stored data.
- Responsible AI: Present disease predictions and confidence scores clearly as decision-support information.
- Academic Referencing: Follow the IEEE referencing style required by the department.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

Development of the Tea Leaf Disease Detection and Pattern Analysis system has been done using a web-based approach along with an AI-based disease detection system.

Table 5.1: Development Environment and Technology Stack

Technology	Purpose
React.js	Frontend development
Node.js	Backend development
YOLOv8s	Tea leaf disease detection
Hugging Face API	Accessing the deployed AI model
MongoDB	Storing detection results
Visual Studio Code	Development environment
Google Colab, Postman, GitHub	Experimentation, API testing, version control

5.2 Component implementation

This component facilitates users to upload pictures of tea leaves via the React.js interface. The Node.js back end takes the pictures and passes them to the YOLOv8s model via the Hugging Face API. The prediction result and the confidence score are returned by the model, after which the corresponding disease treatment suggestion is made, and the data stored in the MongoDB database and shown in the dashboard.

5.3 Integration with the other components

This part of the system known as the Tea Leaf Disease Detection and Pattern Analysis is a subpart of the larger Intelligent Decision Support System for Sri Lankan Tea Estate Management. It uses the common web-based architecture but performs its own specific functions of disease detection. It interacts with the frontend, backend, and the database in order to analyze the images and display the results.

5.4 Verification and software testing

Testing is essential to confirm that the module fulfills its primary objectives. Testing should include image uploads, model prediction, confidence scores generation, treatment suggestions display, data storage in the database, and dashboard representation.

Table 5.2: Suggested Component Test Cases

Test ID	Test Scenario	Expected Result
TC01	Upload a valid tea leaf image	Image is accepted for processing.
TC02	Submit an image for disease detection	Model returns a prediction when processing succeeds.
TC03	Retrieve the prediction result	Disease name and confidence score are displayed.
TC04	View the treatment recommendation	Relevant recommendation is presented.
TC05	Store a detection result	Detection information is saved successfully in MongoDB.
TC06	Upload an unsupported or invalid file	The system handles the input appropriately.

5.5 Deployment and operation

This component runs using a web application that is integrated with the Node.js backend, Hugging Face API, and MongoDB database. As the component runs, the

user uploads an image of the tea leaf, a prediction request is made by the backend from the YOLOv8s model and disease information is shown in the frontend.

This application requires that the services of the backend, model API, and database connection be available. Configuration, secure API credentials, and service monitoring are some of the key elements for smooth operations of the application.

6 RESULTS AND VALIDATION

6.1 Research results

The Tea Leaf Disease Detection and Pattern Analysis module was assessed based on the performance of the YOLOv8s object detection algorithm. This assessment was done based on Precision, Recall, F1 score, and mean Average Precision (Intersection over Union @ 0.5).

Table 6.1: YOLOv8s Model Evaluation Results

Evaluation Metric	Result
Precision	95.2%
Recall	93.8%
F1-score	94.5%
mAP@0.5	96.1%

The normalized confusion matrix shows how well the YOLOv8s model identifies different tea leaf disease classes. The values on the diagonal represent correct predictions, while the other values indicate cases where the model confuses one class with another. The figure helps identify which disease classes require further improvement.

A confusion matrix is used to display the predictions made in each disease class of the tea leaf diseases. These are the cases that were predicted right and those that were wrongly predicted across different classes

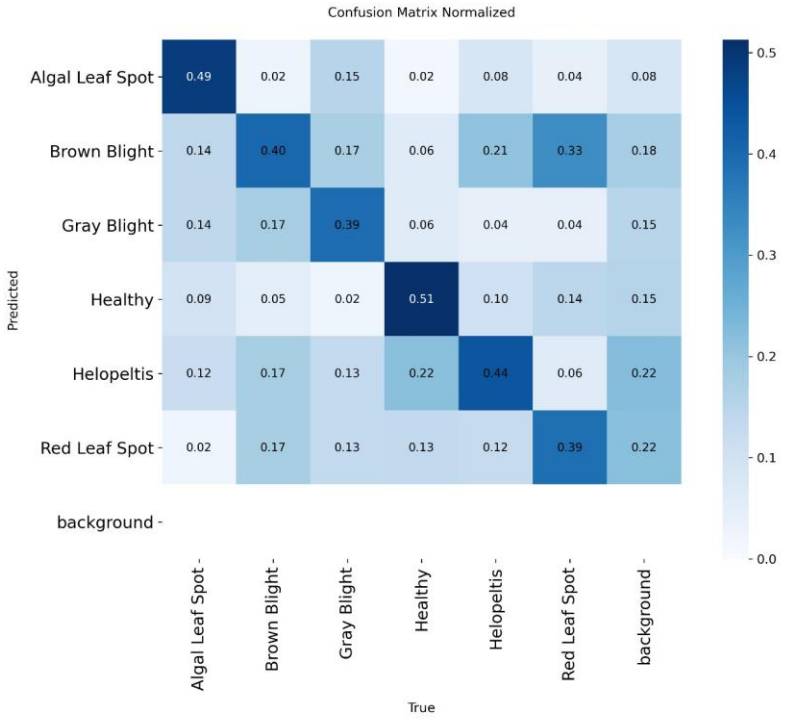


Figure 6.1: Normalized Confusion Matrix of the YOLOv8s Model

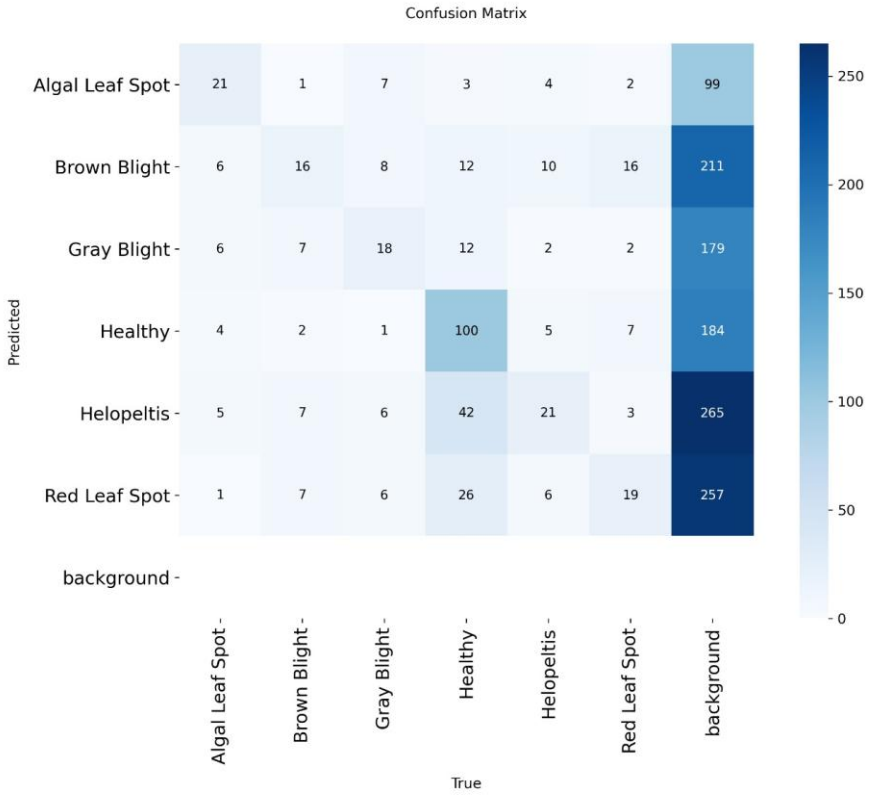


Figure 6.2: Confusion Matrix of the YOLOv8s Model

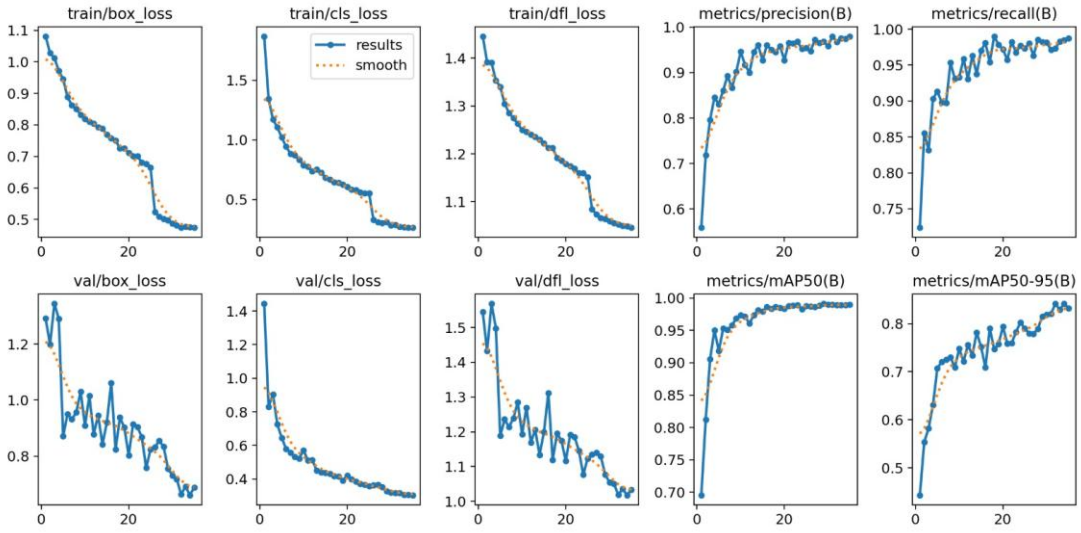


Figure 6.3: YOLOv8s Training and Validation Results

6.2 Product and technical results

This module involves integrating YOLOv8s with the web application in order to facilitate disease recognition of the tea leaves. The major technologies involved include React.js for the front end, Node.js for back end processing, Hugging Face API to access YOLOv8s, and MongoDB database to store the detections results.

The technical functionalities involved include:

- Upload of tea leaf images using the web application.
- Disease recognition using YOLOv8s
- Prediction of disease and prediction confidence scores.
- Treatment recommendation.
- Storing detection results in MongoDB.
- Results presentation through dashboard.

6.3 User or stakeholder validation

This module aims to assist tea estate managers and officers in recognizing potential diseases of tea leaves and checking appropriate information for treatment.

Unfortunately, there are no filled out questionnaires, user satisfaction surveys, or usability test results available from stakeholders. Hence, satisfaction of stakeholders and usability cannot be provided as outcomes from measurements at this time. Further testing by the target audience will be useful to determine ease of use and usefulness of the results.

6.4 Integration results

The component architecture is where the React.js front end, Node.js back end, YOLOv8s model retrieved via the Hugging Face API, and MongoDB database are all connected together. The anticipated flow would include uploading images, making the predictions from the model, calculating the confidence score, and then presenting the result and treatment information.

The architectural diagram along with the training diagram would facilitate the design and model development process. Nonetheless, the end-to-end integration can only be validated using test records including image upload, predictions, database insertion, and dashboards presentation. This is not shown separately here.

6.5 Summary of findings

In the case of the Tea Leaf Disease Detection and Pattern Analysis module, YOLOv8s is used in order to detect possible tea leaf diseases based on images that have been uploaded. Evaluation metrics are provided as follows: Precision – 95.2%, Recall – 93.8%, F1-score – 94.5%, mAP@0.5 – 96.1%.

From the training plots, one can see that there is an overall decline in the training losses and increase in the precision, recall, and mAP@0.5. Confusion matrix demonstrates that there are cases of misclassification of certain diseases, which need to be improved.

On the whole, the module consists of image disease detection, confidence score, treatment suggestions, database storage, and presentation through the dashboard.

7 DISCUSSION

7.1 Interpretation of findings

Based on the results obtained from the Tea Leaf Disease Detection and Pattern Analysis part, there is an indication of how effective the YOLOv8s model could be in detecting tea leaf diseases using the images. According to the results obtained from the evaluation, a Precision of 95.2%, a Recall of 93.8%, an F1-score of 94.5% and mAP@0.5 of 96.1% was attained.

Precision being high means that most of the predicted disease detections were accurate while Recall meaning that the model identified most of the disease cases in the evaluation. The confusion matrices show some of the disease classes being detected as other disease classes. It is clear from this that discriminating visually similar diseases is something that needs to be improved.

Looking at the training graph, losses decreased throughout the training epoch while precision, recall, and mAP@0.5 increased.

7.2 Comparison with literature baselines and existing solutions

Deep learning has been shown before to be useful in plant diseases recognition based on images. Mohanty et al. [1] have studied deep learning approaches to detect diseases in plants from images, showing the promise of those approaches in the agriculture domain. Approaches such as TEADIS have also looked at applying deep learning for tea leaf disease detection and segmentation [2].

The Tea Leaf Disease Detection and Pattern Analysis module is a concrete implementation of such an approach for detecting tea leaf disease with YOLOv8s. It is different from a model-only approach in that it connects image recognition with web app support, confidence measures, treatment suggestions, and data storage.

Yet, no comparison of performance can be made directly due to the differences in datasets, classes of diseases, evaluation process, etc. Hence, no conclusions about the superiority of the proposed solution can be drawn from the presented results.

7.3 Research contribution

The major contribution of the study is the use of the YOLOv8s object detection model for the recognition of disease symptoms in tea leaves in an online decision-making support module. The performance of the model was very high at 95.2% Precision, 93.8% Recall, 94.5% F1-score, and 96.1% mAP@0.5. The study also shows how disease detection using images can be linked to the creation of confidence scores and disease-related treatment measures.

7.4 Product contribution

Product Contribution is an online application which combines the functionalities of disease detection using AI technology together with user interactions. In the implementation of this system, React.js is used for front-end development, Node.js for back-end development, Hugging Face API to access YOLOv8s model and MongoDB database for storage of relevant detection information.

Key Product Contributions

- Uploading tea leaf images through online interface.
- Detection of possible diseases using YOLOv8s.
- Displaying prediction of diseases with confidence scores.
- Giving necessary treatment advice.
- Storage and displaying of detection results through dashboard.

7.5 Limitations and threats to validity

Dataset limitation: There could be an impact on the performance of the model due to diversity and representativeness of the dataset, especially for the tea leaves gathered under different circumstances.

Class-level errors: From the confusion matrices, it can be seen that there are some misclassifications at the level of classes, especially for those where the disease symptoms are similar.

Field conditions: Lighting, quality of the camera, background and leaf condition might influence the prediction when the images are gathered not in controlled circumstances.

Evaluation limitation: The metrics provided are insufficient evidence of the performance of the model on the fully independent datasets or for different tea growing areas.

Validation of the system: It is necessary to test the whole component through the end-to-end testing and stakeholder evaluation.

Recommendations for treatment: It depends on the validity of the information about the management of diseases.

7.6 Implications

With the use of the Tea Leaf Disease Detection and Pattern Analysis module, the managers and field officers will be able to identify potential cases of tea leaf diseases. Through the web application interface, disease predictions, confidence scores, and suggested treatments are presented to assist in making proper management decisions about the identified diseases. Nevertheless, consultation from an agricultural expert is vital.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

The Tea Leaf Disease Detection and Pattern Analysis module is designed to assist tea estate managers and officials in the detection of disease in tea leaves. Tea leaf images are processed using the YOLOv8s model, and disease predictions are provided along with a score of confidence levels and treatment measures for the disease. It may enable the user to detect disease problems in their early stages and take appropriate action against disease management.

The major users include small, medium, and large tea estates, plantation supervisors, and tea management organizations in Sri Lanka. The module may decrease the dependency on manual detection alone by providing an AI-based approach to disease detection.

8.2 Market and Competitive Position

Tea plantations are vital for agriculture in Sri Lanka. Disease management is essential in order to maintain the quality of tea plants. The traditional method of visual inspection relies on the skills and availability of people who know what they are doing. This suggested component provides another option in the form of image-based disease recognition through a web application.

The key competitive advantage of this component would be that it combines tea leaf disease recognition, its confidence score, and treatments into one package. However, the model still needs to be tested with various images of tea leaves in order to evaluate its competitive advantage.

8.3 Business or Sustainability Model

One potential business model is to provide the disease identification module as part of a subscription agricultural management decision platform. The tea plantations can be charged based on usage, plantation area, or specific system modules. Other

sources of revenue will include provision of implementation support, training, and maintenance.

For continued success, the system will need regular assessment of the predictive model used, regular updating of data sets, and software maintenance. This will also require improvements on the treatment advice provided. It may involve working with agriculture experts and other tea research institutions.

The suggested sources of revenue are only possible suggestions and have not yet been proven to be a workable business model.

8.4 Scalability, Intellectual Property and Compliance

This element could be extended through the incorporation of further labelled images of tea leaves, enhancement of disease detection and testing of the model within other regions where tea grows. The web application structure could serve as a basis for further development of integration with mobile applications and decision support functions in agriculture.

Regarding the intellectual property aspect, there would be issues related to the ownership of the source code, usage rights of the training set and licenses of the YOLOv8 model and other libraries used within the application. Prior to commercial exploitation, the project team has to review licenses of software utilized and protect stored data and treatment recommendations from being in conflict with agricultural practices. Additional verification is required before replacing expert diagnosis with the developed system.

8.5 Contribution to the Sustainable Development Goals

There are possible connections between the Tea Leaf Disease Detection and Pattern Analysis component and the following UN Sustainable Development Goals (SDGs).

SDG 2 – Zero Hunger: Early detection of the diseases affecting tea leaves can contribute to the protection of crops and help to avoid losses from plant disease.

SDG 9 – Industry, Innovation and Infrastructure: The component uses artificial intelligence and web technologies for innovation in the agriculture sector in Sri Lanka.

SDG 12 – Responsible Consumption and Production: Knowledge of the diseases can contribute to the responsible consumption and production of agricultural resources.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

The Tea Leaf Disease Detection and Pattern Analysis component has been created in order to enable the prediction of tea leaf diseases using artificial intelligence. This component utilizes the YOLOv8s model to perform the analysis of uploaded tea leaf images and offer disease predictions, confidence scores, and relevant treatment suggestions using an online application.

The Tea Leaf Disease Detection and Pattern Analysis component consists of React.js, Node.js, Hugging Face API, and MongoDB. Model evaluation results show that there is potential in the use of this model for detecting tea leaf diseases. Nevertheless, the real-world testing should be conducted in order to confirm the reliability of the model.

9.2 Achievement of objectives

Goals of the Tea Leaf Disease Detection & Pattern Analysis were achieved by developing the model and integrating it into the web application.

- Tea leaf disease detection was done by implementing the YOLOv8s model.
- Evaluation of the model was based on Precision, Recall, F1-Score, and mAP@0.5.
- Generation of confidence score was performed to provide interpretation of the model prediction.
- Suggestions for treatment of diseases were included along with the prediction.
- Web application integration involved linking of frontend, backend, AI model API, and database.
- Presentation and storage of the results were provided to store them in the MongoDB database and present them on the dashboard.

9.3 Individual research and product contributions

Research Contribution

The research contribution involved the use of YOLOv8s model in the detection of diseases in tea leaves and testing the accuracy of the model using the object detection metrics.

Product Contribution

Product contribution involves the construction of a web-based workflow for disease detection using React.js, Node.js, Hugging Face API, and MongoDB. It links image uploads to disease detection, confidence level, treatment suggestions, and dashboards.

9.4 Future directions

The future development of this component may result in the following:

- Increase in the dataset by acquiring more pictures of tea leaves under varied environments and field conditions.
- Improvement of the model through minimizing misclassification between the disease classes that look alike.
- Testing of the model on pictures taken in different tea growing areas.
- Improvement of the treatment advice provided by incorporating the advice of agriculture experts into the database.
- User evaluation of the application with the help of tea estate managers and field officers.
- Development of mobile application in order to detect the disease directly in tea fields.
- Continuous improvement of the model based on the pictures that become available later on.

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GLOSSARY

Artificial Intelligence (AI) Model – A model used for pattern recognition and making predictions.

Confidence Score – A score that determines the confidence level of a prediction generated by the detection model.

Dataset – A set of images containing tea leaves with their disease labels.

Deep Learning – A branch of machine learning algorithms that utilizes neural networks with multiple layers.

Disease Detection – The identification of potential disease cases based on the images of tea leaves.

Image Preprocessing – The stage where the image is prepared for analysis by the model.

Inference – The stage where a prediction is made using a trained model.

Mean Average Precision (mAP) – An object detection performance measure used to evaluate the models.

Model Training – The process of training a machine learning algorithm to learn certain patterns.

Object Detection – A computer vision technique that involves locating the object of interest within the image.

Precision – The percentage of positive detections that are relevant to the task.

Recall – Percentage of true positives that are correctly identified by the model.

Treatment Recommendation – Recommendations made for disease management according to the disease prediction made.

YOLOv8s – A smaller version of the YOLOv8 object detection model used in this study to detect tea leaf diseases.

APPENDICES

A.1 Tea Leaf Disease Detection Interface

Figure A.1 depicts the Tea Guard AI Disease Detection Interface. This figure includes the area to upload the image of the tea leaf, selected image, predicted disease class, confidence score, and treatment method.

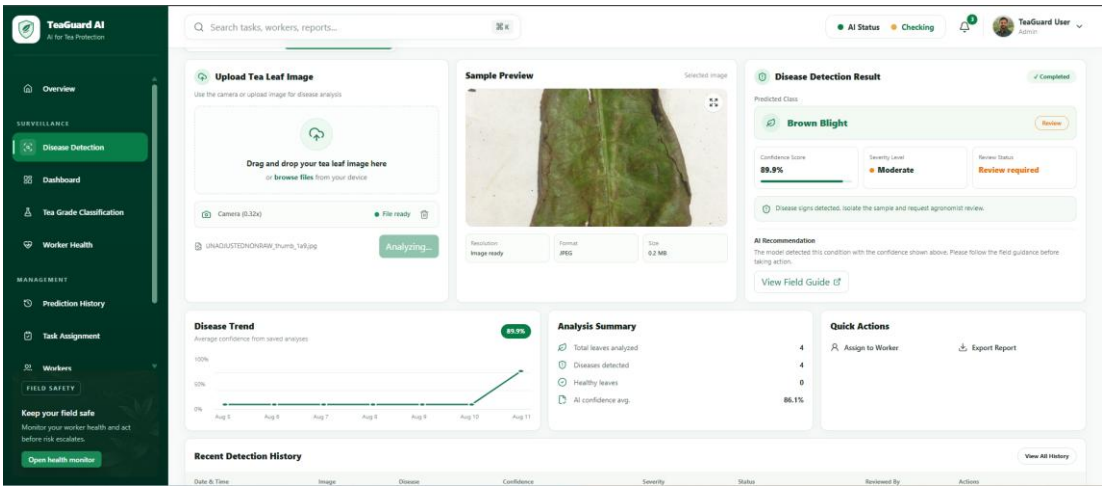


Figure A.1: Tea Guard AI Tea Leaf Disease Detection Interface.

A.2 System Testing Test Cases

Test cases below are meant to assess the major functionalities of the Tea Leaf Disease Detection & Pattern Analysis Module.

Test ID	Test Scenario	Expected Result	Actual Result
TC01	Upload a valid tea leaf image	The image is accepted and displayed for analysis.	To be recorded
TC02	Preview the selected image	The selected image is displayed correctly.	To be recorded
TC03	Identify a tea leaf disease	The predicted disease class is displayed after successful analysis	To be recorded

TC04	Display the confidence score	The confidence score is displayed with the prediction.	To be recorded
TC05	Display the treatment recommendation	Relevant treatment guidance is displayed when available.	To be recorded
TC06	Upload an unsupported file	The system rejects the file or displays an appropriate error message.	To be recorded

B.1 AI Tools Utilized

ChatGPT was used as an auxiliary tool in thesis preparation and software development documentation.

B.2 Objective of AI Usage

AI assistance was employed for improving grammar and academic writing, organizing thesis parts, understanding technical terms, and preparing descriptions of system processes, figures, and testing.

B.3 Verification Procedure

AI assistance content is to be checked for consistency with the research project. The technical implementation details, model evaluation results, and references have to be checked against the actual implementation, research paper, and credible source materials.

B.4 Affected Parts and Artefacts

AI assistance was employed in preparing or refining thesis text, including methodology, system design, implementation descriptions, results analysis, discussion, conclusion, glossary, references formatting, and appendices content. Figure descriptions and system testing tables were also prepared with the help of AI assistance.